

Non-Random Missingness in Anthropometric Measurements: A Multi-Country Analysis of South-Asian DHS Data (2011–2022)

Submitted by

Tahsina Akter

Roll No: SK-182-033

Session: 2024–2025



Supervised by

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Professor

ISRT, University of Dhaka

**A Project Report Submitted in Partial Fulfillment
of the Requirements for the Degree of Bachelor of Science (Honors) in Applied Statistics**

Institute of Statistical Research and Training (ISRT)

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Declaration

I hereby declare that the project entitled “*Non-Random Missingness in Anthropometric Measurements: A Multi-Country Analysis of South Asian DHS Data (2011–2022)*” is my own original work completed under the supervision of **Dr. Md Hasinur Rahaman Khan**, Professor, ISRT, University of Dhaka.

All sources of information used in this study have been duly acknowledged and cited. This work has not been submitted to any other institution for the award of any degree or diploma.

Tahsina Akter

Signed:

Date:

Dedicated to my parents and all my loved ones.

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Finally, I acknowledge the Demographic and Health Surveys (DHS) Program for granting access to the datasets used in this study.

Abstract

Height-for-age Z-scores (HAZ) from all Demographic and Health Surveys (DHS) are widely used to monitor child undernutrition and assess socioeconomic gradients in child growth. However, anthropometric related datasets often contain missing or biologically implausible values that are commonly excluded without formal evaluation of the underlying missingness structure. This study examines whether child height usability in pooled South Asian DHS data is randomly distributed or systematically associated with observable demographic and socioeconomic characteristics.

Using 11 DHS surveys conducted between 2011 and 2022 in Bangladesh, India, Pakistan and Nepal, a harmonized pooled dataset of 0–59 months of children was assembled. A binary indicator of height usability was defined based on the presence of observed HAZ scores within the plausibility range recommended by the World Health Organization (WHO). Survey-weighted logistic regression models were used to identify predictors of height usability, incorporating characteristics at the child, mother and household level. Temporal variation was assessed through stratified models and interaction terms. To evaluate inferential implications, survey weighted linear regression models of HAZ were estimated using both complete case analysis and multiple imputation with predictive mean matching.

Results indicate that height usability is systematically structured rather than randomly distributed. Younger children are disproportionately represented among unusable measurements and maternal education, household wealth and residence are associated with variation in the likelihood of usable measurement. Temporal analyses show modest improvements in overall usability across survey waves. Although changes are not uniform across socioeconomic strata, inclusion of country and survey-period indicators does not materially alter key associations.

Despite evidence of structured missingness, comparisons between complete case and multiply imputed models demonstrate substantial stability in estimated determinants of child HAZ. These findings suggest that while anthropometric missingness in pooled DHS data is socially patterned, its impact on multivariable nutritional inference may be limited when relevant covariates are appropriately modeled. The study underscores the importance of explicitly evaluating missing data mechanisms in secondary analyses of complex survey data to enhance methodological transparency and inferential credibility.

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List of Abbreviations

DHS Demographic and Health Surveys

HAZ Height-for-Age Z-score

WHO World Health Organization

SD Standard Deviation

OR Odds Ratio

Adj. OR Adjusted Odds Ratio

CI Confidence Interval

PSU Primary Sampling Unit

MCAR Missing Completely at Random

MAR Missing at Random

MNAR Missing Not at Random

MI Multiple Imputation

PMM Predictive Mean Matching

SDGs Sustainable Development Goals

Chapter 1

Introduction

1.1 Background of the Study

Missing data are pervasive in empirical research and represent one of the most challenging problems in statistical analysis. When observations are incomplete, standard estimation procedures may yield biased and inefficient results unless the underlying missing data mechanism is properly understood and addressed (Rubin, 1987). The modern theoretical foundation of missing data analysis was established by Rubin (1976), who formalized a probabilistic framework for classifying missingness mechanisms and clarifying their implications for statistical inference.

Rubin's framework distinguishes among Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR). Under MCAR, the probability that an observation is missing is independent of both observed and unobserved data, implying that complete-case analysis produces unbiased parameter estimates, although at the cost of reduced statistical efficiency. Under MAR, missingness may depend on observed characteristics but not on the unobserved value itself, allowing likelihood based and multiple imputation approaches to recover unbiased estimates when models are correctly specified. In contrast, MNAR arises when missingness depends directly on unobserved values, creating identification challenges and typically requiring stronger assumptions for valid inference (Little and Rubin, 2002; Rubin, 1976; Schafer, 1997).

Failure to account for systematic missingness can distort parameter estimates, inflate standard errors, and ultimately lead to misleading substantive conclusions (Little and Rubin, 2002; Graham, 2009). Despite substantial methodological advancements, many applied studies continue to rely on complete-case analysis without formally evaluating whether the MCAR assumption is plausible (Enders, 2010). Such practices are particularly concerning in public health research, where policy decisions frequently depend on accurate prevalence estimation and the identification of vulnerable populations. Among various domains of public health data, anthropometric measurements are especially susceptible to issues of non-response, measurement error and biological implausibility (Uliaszek and Kerr, 1999; Pullum, 2013).

These methodological concerns are particularly salient in large-scale population health surveys, where measurement completeness directly affects the validity of prevalence estimates and inequality assessments. In such settings, missingness is rarely a purely statistical inconvenience; rather, it may reflect operational, socioeconomic, or behavioral processes embedded within data collection environments. Understanding whether incomplete observations arise randomly or systematically is therefore essential when survey-based

indicators serve as the foundation for public health monitoring and policy evaluation ([Heeringa et al., 2017](#); [Victora et al., 2008](#)).

Anthropometric indicators, particularly height-for-age Z-scores (HAZ), are central to assessing chronic undernutrition and stunting among children under five years of age. HAZ is calculated using the Child Growth Standards of the World Health Organization (WHO) and reflects cumulative growth deficits relative to a healthy reference population ([World Health Organization, 2006](#)). Stunting remains a major global health concern and reliable estimation of HAZ distributions is essential to monitor nutritional progress and inform interventions ([UNICEF et al., 2023](#); [Black et al., 2008](#); [Bhutta et al., 2013](#)).

However, anthropometric data collected in large scale surveys are vulnerable to missingness and measurement challenges. Height measurements may be unavailable due to child absence during survey visits, caregiver refusal, equipment malfunction or enumerator error ([Pullum, 2013](#); [Croft et al., 2018](#)). In addition, extreme anthropometric values may arise from inaccuracies in the recording or errors in data entry. To address data quality concerns, the WHO recommends excluding HAZ values outside the biologically plausible range of $[-6, +6]$ ([World Health Organization, 2006](#)). Although such exclusion can improve reliability, it can also introduce selection bias if missing or implausible observations are systematically associated with particular demographic or socioeconomic characteristics ([Hernán et al., 2004](#); [Cole et al., 2010](#)).

The Demographic and Health Surveys (DHS) Program is one of the main sources of nationally representative health and nutrition data in low and middle income countries ([The Demographic and Health Survey \(DHS\) Program, 2022](#)). DHS surveys employ standardized sampling frameworks, calibrated measurement tools, and harmonized data collection procedures to ensure cross-country and temporal comparability ([Croft et al., 2018](#)). Anthropometric modules within DHS surveys provide critical information for estimating child stunting and evaluating disparities across regions and subgroups of population.

Although DHS surveys follow standardized protocols, the social, economic, and logistical contexts in which data collection occurs differ substantially across regions. Variations in fieldwork conditions, household accessibility, and socioeconomic structure may influence both the completeness and quality of anthropometric measurements. Consequently, the structure of missingness may not be uniform across geographic settings, underscoring the importance of region specific evaluation within otherwise harmonized survey systems ([Croft et al., 2018](#); [Pullum, 2013](#)).

Nevertheless, evidence suggests that anthropometric data quality may vary across socioeconomic strata and geographic contexts ([Pullum, 2013](#)). If missing or implausible height measurements are disproportionately concentrated among certain groups, such as younger children, children of less educated mothers, or households of particular wealth levels, then analyses restricted to usable observations may produce biased estimates of stunting prevalence and its determinants ([Little and Rubin, 2002](#)). Therefore, a systematic evaluation of anthropometric missingness is essential before drawing substantive conclusions about child nutritional outcomes.

The issue is particularly relevant in South Asia, a region that has historically experienced high burdens of child undernutrition despite notable economic progress ([UNICEF et al., 2023](#)). Although national averages of stunting have declined in recent years, substantial socioeconomic and rural-urban disparities persist across South Asian countries, including Bangladesh, India, Pakistan, and Nepal ([UNICEF et al., 2023](#); [World Bank, 2023](#); [Subramanian et al., 2009](#); [Fotso, 2006](#)). These countries have implemented multiple DHS survey rounds between 2011 and 2022, producing harmonized anthropometric data that are

comparable across time and settings ([The Demographic and Health Survey \(DHS\) Program, 2022](#); [Croft et al., 2018](#)). Pooling repeated cross-sectional surveys increases statistical precision and facilitates the examination of temporal trends in nutritional outcomes and data quality ([Heeringa et al., 2017](#)).

However, if anthropometric missingness differs systematically across countries, survey periods, or socioeconomic groups, pooled analyses may mask underlying heterogeneity and generate biased or misleading inferences ([Little and Rubin, 2002](#); [Rubin, 1987](#)). In large scale health surveys, failure to evaluate the structure of missing data mechanisms can compromise both prevalence estimation and regression-based inference ([Schafer, 1997](#)). Given the policy importance of accurate stunting estimation in South Asia, assessing whether missing or biologically implausible anthropometric measurements occur randomly or systematically is therefore not merely a technical exercise but a prerequisite for credible statistical inference ([World Health Organization, 2006](#); [UNICEF et al., 2023](#)). If particular subgroups of children are disproportionately represented among those with unusable height records, standard complete case approaches may yield biased estimates of both prevalence and determinants of chronic undernutrition. Accordingly, rigorous evaluation of anthropometric missingness within DHS data is essential to ensure the validity and reliability of empirical conclusions in child nutrition research.

1.2 Research Gap and Rationale

A substantial body of literature has used DHS data to estimate the prevalence of stunting and examine its socioeconomic determinants across low and middle income countries. These studies typically rely on HAZ scores derived from WHO growth standards and employ regression based methods to identify correlates of child undernutrition ([Victora et al., 2008](#); [Headey et al., 2016](#); [Hong, 2006](#)). Although such analyses have contributed significantly to understanding child growth disparities, they often treat missing or biologically implausible anthropometric observations as a purely technical nuisance rather than a substantive analytical issue.

In most applied nutrition studies, observations with missing HAZ values or implausible measurements are excluded before analysis, and subsequently complete case methods are implemented without formally assessing the underlying missingness mechanism ([Sterne et al., 2009](#)). Although the exclusion of implausible values is recommended for data quality purposes ([World Health Organization, 2006](#)), relatively few studies investigate whether such exclusions disproportionately affect specific demographic or socioeconomic subgroups. When missingness or implausibility is systematically associated with observable characteristics, such as child age, maternal education, household wealth, or geographic location, complete case estimates may be biased and may distort inference regarding inequality in child growth outcomes ([Little and Rubin, 2002](#)).

Methodological work has long emphasized the importance of evaluating missing data mechanisms before proceeding with substantive analysis ([Little and Rubin, 2002](#); [Rubin, 1987](#)). However, empirical investigations of anthropometric missingness within DHS data remain limited, particularly in multi-country settings. Evidence from Africa suggests that missingness of height data in DHS surveys is not random, but may have a limited impact on stunting prevalence estimates, thus motivating region specific evaluations in South Asia ([Finaret and Hutchinson, 2018](#)). Although demographic structure, survey implementation contexts and socioeconomic gradients differ across regions and findings from sub-Saharan Africa cannot be assumed to generalize to South Asia. Existing research often focuses either on single-country analyses

or general diagnostics of data quality, without examining pooled cross national patterns or temporal variation in anthropometric data completeness (Pullum, 2013; Croft et al., 2018).

Moreover, while multiple imputation techniques are widely recommended under the Missing at Random (MAR) assumption (Rubin, 1987), their application in DHS based anthropometric research has not been consistently integrated with formal assessments of missingness structure (White et al., 2011; Carpenter and Kenward, 2013). As a result, there remains a gap between methodological recommendations and routine applied practice in large scale nutrition studies.

This study addresses these limitations by systematically evaluating the prevalence, predictors, and temporal evolution of missing and implausible child height measurements across eleven DHS surveys conducted between 2011 and 2022 in four South Asian countries. By combining survey weighted descriptive analysis, multivariable logistic regression, interaction models, and multiple imputation techniques, the study provides an integrated assessment of anthropometric data quality and its implications for inference. In doing so, it contributes empirical evidence on whether standard complete case approaches may introduce bias in the estimation of child nutritional outcomes within a region of high policy relevance (Bhutta et al., 2013).

1.3 Objectives of the Study

The overarching objective of this study is to determine whether missing and biologically implausible height-for-age Z-score (HAZ) measurements in pooled South Asian Demographic and Health Surveys (DHS) data occur randomly or exhibit systematic associations with observable child, maternal, and household characteristics.

To achieve this aim, the study pursues the following specific objectives:

- i) To estimate the prevalence of missing and biologically implausible child height measurements across eleven DHS survey rounds conducted between 2011 and 2022 in Bangladesh, India, Pakistan, and Nepal.
- ii) To examine the association between height usability and child, maternal, and household characteristics using survey-weighted multivariable logistic regression models.
- iii) To assess whether patterns of height usability differ across survey periods and socioeconomic groups through interaction analyses.
- iv) To evaluate the inferential implications of complete-case analysis by comparing regression estimates of child HAZ obtained from complete-case and multiple imputation approaches using Predictive Mean Matching (PMM).

1.4 Organization of the Report

The remainder of this report is structured as follows. Chapter 2 outlines the methodological framework, including the evaluation of missingness mechanisms, survey-weighted regression modeling, interaction analyses, and the implementation of multiple imputation using Predictive Mean Matching (PMM). Chapter

3 describes the data sources, sample construction, variable definitions, and survey design considerations underlying the pooled DHS dataset.

Chapter 4 presents the empirical findings, including descriptive statistics, multivariable regression results, interaction effects across survey periods and socioeconomic groups, and comparisons between complete-case and imputed estimates. Chapter 5 discusses the implications of the findings in light of existing literature and methodological considerations, then concludes the study by summarizing key insights, highlighting limitations, and suggesting directions for future research. Finally, references cited throughout the report are compiled at the end, followed by an Appendix containing supplementary tables, figures, and key R code used in the analysis.

Chapter 2

Methodology

2.1 Overview of the Analytical Framework

This study adopts a structured statistical framework to evaluate systematic patterns in anthropometric usability and assess their implications for statistical inference in survey-based child nutrition research. The analytical strategy consists of four principal components: (i) descriptive evaluation of data completeness under a complex survey design, (ii) multivariable modeling of a binary outcome using survey-weighted logistic regression, (iii) assessment of interaction effects and marginal predicted probabilities across subgroups and time periods, and (iv) implementation of multiple imputation to address incomplete observations under plausible missing data mechanisms.

All statistical analyses are conducted within a design-based inference framework to account for stratification, clustering, and unequal sampling probabilities inherent in multistage household surveys (Heeringa et al., 2017; Cochran, 1977). Model estimation and variance calculation are performed using survey-adjusted procedures to ensure unbiased point estimates and valid standard errors.

2.2 Complex Survey-Design and Weighted Inference

Large scale DHS surveys typically employ multistage stratified cluster sampling designs to produce nationally representative estimates (Heeringa et al., 2017; Lohr, 2019). In such designs, primary sampling units (PSUs) are selected within strata defined by geographic and demographic characteristics, followed by systematic selection of households within clusters. Because observations are not obtained through simple random sampling, conventional statistical methods that ignore survey-design may yield biased standard errors and misleading inference.

Let w_i denote the sampling weight associated with observation i , reflecting the inverse probability of selection adjusted for non-response. DHS sampling weights were rescaled by dividing by 1,000,000 in accordance with DHS guidelines prior to survey declaration. Weighted estimators are constructed as

$$\hat{\theta}_w = \frac{\sum_{i=1}^n w_i y_i}{\sum_{i=1}^n w_i},$$

where y_i represents the variable of interest, w_i is the sampling weights, and $i = 1, \dots, n$ indexes sampled individuals. Variance estimation under complex survey-designs commonly relies on Taylor series

linearization or replication-based methods to account for clustering and stratification (Heeringa et al., 2017). Design-based inference ensures that both point estimates and confidence intervals appropriately reflect the sampling structure (Korn and Graubard, 1999). The pooled dataset was declared as survey data specifying PSU, strata, and sampling weights prior to analysis.

2.3 Survey-Weighted Descriptive and Bivariate Analysis

Descriptive statistics were computed using survey-weighted procedures to account for stratification, clustering, and unequal sampling probabilities. Weighted proportions and means were estimated for categorical and continuous variables, respectively.

Bivariate associations between height usability and categorical covariates were assessed using design-based Pearson chi-square tests adjusted for complex survey-design. For continuous variables, differences in weighted means were evaluated using survey-adjusted t-tests based on Taylor series linearization. These methods ensure valid variance estimation and hypothesis testing under multistage sampling frameworks (Heeringa et al., 2017; Korn and Graubard, 1999).

2.4 Evaluation of Missing Data Mechanisms

The statistical consequences of incomplete data are critically dependent on the underlying missing mechanism. Following the framework introduced by Rubin (1976), missing data mechanisms are classified as Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR).

Let R_i denote an indicator variable equal to 1 if observation i is observed and 0 otherwise, and let Y_i denote the variable subject to missingness, with X_i representing fully observed covariates. Under MCAR,

$$P(R_i = 1 \mid Y_i, X_i) = P(R_i = 1),$$

implying independence between missingness and both observed and unobserved data. Under MAR,

$$P(R_i = 1 \mid Y_i, X_i) = P(R_i = 1 \mid X_i),$$

indicating that missingness may depend on observed covariates but not on the unobserved value itself. Under MNAR, missingness depends directly on the unobserved outcome.

Empirical evaluation of missingness patterns often involves modeling response indicators as functions of observed covariates (Little and Rubin, 2002; Enders, 2010). Evidence of systematic association between missingness and observed characteristics suggests departure from MCAR and may be consistent with the MAR assumption, conditional on measured variables. It is important to note that distinguishing between MAR and MNAR cannot be achieved solely through observed data, and therefore conclusions regarding the missingness mechanism remain conditional on measured covariates.

In the context of this study, the response indicator corresponds to the usability of height-for-age Z-score measurements, allowing formal evaluation of whether unusable anthropometric observations occur randomly or are systematically associated with observable child, maternal, or household characteristics.

2.5 Survey-Weighted Logistic Regression

When the response variable is binary, logistic regression provides an appropriate and widely used modeling framework (Hosmer et al., 2013). Let Y_i denote a binary outcome taking values 1 for the event of interest and 0 for the reference category. For each individual i , the model specifies the conditional probability of the event as

$$P(Y_i = 1 \mid \mathbf{X}_i) = \pi_i.$$

The relationship between the covariates and the outcome is modeled through the logit link function,

$$\text{logit}(\pi_i) = \log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \boldsymbol{\beta}^\top \mathbf{X}_i,$$

where π_i denotes the probability that the outcome equals 1 given the covariates, $\mathbf{X}_i = (X_{i1}, X_{i2}, \dots, X_{ip})^\top$ is a $p \times 1$ vector of explanatory variables, $\boldsymbol{\beta} = (\beta_1, \beta_2, \dots, \beta_p)^\top$ is the corresponding vector of regression coefficients, and $i = 1, \dots, n$ indexes sampled individuals.

The intercept β_0 represents the log odds of the outcome when all covariates are at their reference or baseline levels, that is, zero for continuous predictors and the designated reference category for categorical predictors. Thus, $\exp(\beta_0)$ corresponds to the baseline odds of the event.

For a given covariate X_j , the coefficient β_j reflects the expected change in the log odds of the outcome associated with a one unit increase in X_j , holding all other variables constant, where $j = 1, \dots, p$. Exponentiating the coefficient yields $\exp(\beta_j)$, which can be interpreted as the odds ratio; the factor by which the odds of the outcome change for a one unit increase in X_j , relative to the baseline, conditional on the remaining covariates. Values greater than one indicate higher odds, whereas values less than one indicate lower odds.

When data arise from complex survey designs, estimation proceeds via weighted pseudo-maximum likelihood to account for unequal selection probabilities. Sampling weights are incorporated to produce population-representative estimates, and design features such as clustering and stratification are explicitly accommodated. Variance estimation relies on design-based methods, ensuring valid statistical inference under multistage sampling schemes (Heeringa et al., 2017).

2.6 Interaction Effects and Marginal Predictions

To examine whether associations differ across subgroups or time periods, interaction terms can be incorporated into the logistic regression framework. For two covariates X_1 and X_2 , the model with an interaction term is specified as

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 (X_{1i} \times X_{2i}),$$

where π_i denotes the conditional probability of the outcome for the individual i and $i = 1, \dots, n$ indexes sampled individuals. Here, X_{1i} and X_{2i} represent the primary explanatory variables of interest, and $(X_{1i} \times X_{2i})$ is their product term. The intercept β_0 corresponds to the log odds of the outcome when all covariates are at their baseline or reference levels.

The coefficients β_1 and β_2 represent the associations between each covariate and the log odds of the

outcome when the interacting variable equals its reference value. The interaction coefficient β_3 captures effect modification, indicating the extent to which the association between X_1 and the outcome changes with the level of X_2 (and vice versa). Therefore, a statistically significant β_3 provides evidence that the effect of one covariate depends on the other.

Because coefficients in nonlinear models are expressed on the log odds scale, they do not directly translate into changes in predicted probabilities. For substantive interpretation, predicted probabilities are computed as

$$\hat{\pi}_i = \frac{\exp(\hat{\eta}_i)}{1 + \exp(\hat{\eta}_i)},$$

where

$$\hat{\eta}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \hat{\beta}_3 (X_{1i} \times X_{2i}).$$

Here, $\hat{\eta}_i$ denotes the estimated linear predictor for observation i , and $\hat{\beta}$ represents the estimated regression coefficients.

Predicted marginal probabilities are obtained using post-estimation marginal standardization, whereby predictions are averaged over the observed covariate distribution. Graphical presentation of these adjusted predictions facilitates interpretation of interaction effects in nonlinear models by expressing results on the probability scale rather than the log-odds scale.

2.7 Survey-Weighted Linear Regression for Continuous Outcomes

For analyses involving continuous dependent variables, survey-weighted linear regression models were estimated. Let Y_i denote a continuous outcome for observation i . The model is specified as

$$Y_i = \alpha + \boldsymbol{\beta}^\top X_i + \varepsilon_i,$$

where α is the intercept, representing the expected value of Y when all covariates are at their reference or baseline levels, $X_i = (X_{i1}, \dots, X_{ip})^\top$ is a vector of explanatory variables for individual i , $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^\top$ is the corresponding vector of regression coefficients and ε_i is the error term, assumed to have mean zero and finite variance. Each coefficient β_j represents the expected change in the outcome associated with a one-unit increase in X_{ij} , holding other variables constant.

Estimation was conducted using survey adjusted procedures to account for unequal selection probabilities and the multistage sampling design. Sampling weights were incorporated to obtain population-representative point estimates, and variance estimation relied on Taylor series linearization to produce valid standard errors under clustering and stratification ([Heeringa et al., 2017](#)).

2.8 Multiple Imputation and Predictive Mean Matching

When data are plausibly Missing at Random (MAR), multiple imputation (MI) provides a principled approach for handling incomplete observations ([Rubin, 1987](#)). MI replaces each missing value with $m > 1$ simulated values drawn from an appropriate predictive distribution, producing m completed datasets, where m is the number of imputed datasets to generate. Although the MAR assumption cannot be empirically verified and MNAR mechanisms cannot be entirely ruled out, multiple imputation provides a

principled sensitivity framework when missingness is conditionally associated with observed characteristics. Each completed dataset is analyzed separately, yielding parameter estimates $\hat{\theta}_k$ and associated variances U_k for $k = 1, \dots, m$. Final pooled estimates are obtained using Rubin's rules:

$$\bar{\theta} = \frac{1}{m} \sum_{k=1}^m \hat{\theta}_k,$$

$$T = \bar{U} + \left(1 + \frac{1}{m}\right) B,$$

where

$$\bar{U} = \frac{1}{m} \sum_{k=1}^m U_k$$

is the within-imputation variance and

$$B = \frac{1}{m-1} \sum_{k=1}^m (\hat{\theta}_k - \bar{\theta})^2$$

is the between-imputation variance ([Rubin, 1987](#)).

Predictive Mean Matching (PMM) is a semi-parametric imputation method in which missing values are replaced with observed values from donors having similar predicted means ([Van Buuren, 2018](#)). PMM preserves the empirical distribution of the data and reduces sensitivity to normality assumptions, making it particularly suitable for continuous variables.

Convergence diagnostics, trace plots, and distributional comparisons between observed and imputed values are commonly evaluated to assess the adequacy and stability of the imputation procedure ([Van Buuren, 2018](#)).

Chapter 3

Data and Variables

3.1 Data Sources and Study Population

This study draws on nationally representative data from eleven Demographic and Health Survey (DHS) rounds conducted between 2011 and 2022 in four South Asian countries: Bangladesh (2011, 2014, 2017–18, 2022), India (2015–16, 2019–21), Pakistan (2012–13, 2017–18), and Nepal (2011, 2016, 2022) ([National Institute of Population Research and Training \(NIPORT\), 2023](#); [International Institute for Population Sciences \(IIPS\), 2021](#); [National Institute of Population Studies \(NIPS\), 2019](#); [Ministry of Health and Population \(MOHP\), 2023](#)). These countries were selected due to their sustained burden of child undernutrition and the availability of multiple harmonized DHS rounds within the past decade.

The analysis utilizes the Children’s Recode (KR) files from each survey. These files contain detailed information on children aged 0–59 months at the time of the survey, including anthropometric measurements, maternal characteristics, and household-level socioeconomic indicators. The study population includes children aged 0–59 months at the time of survey, consistent with WHO definitions of stunting and global nutrition monitoring standards.

All eleven survey datasets were appended to create a pooled cross-national dataset. Country identifiers and survey-period indicators were retained to preserve round-specific distinctions and enable temporal comparisons.

3.2 Harmonization and Survey-Design Adjustment

DHS surveys are independently implemented across countries and survey rounds. Consequently, primary sampling unit (PSU) and stratum identifiers are not globally unique. Directly appending datasets may therefore result in identical numeric codes representing distinct sampling units across countries.

To preserve the integrity of design-based inference, new PSU and stratum identifiers were constructed by concatenating country identifiers with original cluster and stratum variables. This ensured that each sampling unit remained uniquely identifiable in the pooled dataset and prevented artificial merging of survey clusters.

Sampling weights were derived from the DHS individual sample weight variable `v005`, which represents women’s individual weights recorded with six decimal places, and were rescaled by dividing by 1,000,000

in accordance with DHS guidelines. The application of survey weights ensures that estimates reflect the sampling design and are nationally representative within each country and survey round. Because surveys were pooled across countries without reweighting to population size, pooled estimates reflect the combined analytic sample rather than population-weighted regional aggregates. The pooled dataset was declared as survey data using reconstructed PSU identifiers, stratification variables, and probability weights. Variance estimation was conducted using Taylor series linearization, with single-unit strata centered to maintain numerical stability in strata containing only one PSU.

3.3 Construction of the Outcome Variable

The primary outcome of interest is the usability of child HAZ scores. In addition, HAZ scores are analyzed as a continuous outcome in the HAZ determinant models. In DHS surveys, HAZ values are derived from the anthropometric variable `hw70`, which represents the height-for-age standard deviation calculated according to the 2006 WHO Child Growth Standards (World Health Organization, 2006; De Onis et al., 2007). The variable `hw70` is stored as an integer scaled by 100 in the DHS recode files (ICF, 2018); therefore, conventional Z-scores were obtained by dividing `hw70` by 100.

Biologically implausible HAZ values were identified using the WHO-recommended plausibility range of $[-6, +6]$ for height-for-age Z-scores (World Health Organization, 2006). Observations falling outside this range were classified as implausible. In addition, cases with missing HAZ values were identified using the original DHS survey coding (ICF, 2018).

A binary indicator variable, termed *height usability*, was constructed such that observations were coded as 1 if the HAZ value was present and biologically plausible, and 0 if the value was either missing or implausible. The indicator is formally defined as:

$$H_i = \begin{cases} 1, & \text{if } -6 \leq Z_i \leq 6 \text{ and } Z_i \text{ is observed,} \\ 0, & \text{if } Z_i \text{ is missing or } |Z_i| > 6, \end{cases}$$

where Z_i denotes the height-for-age Z-score of child i . The interval $[-6, 6]$ corresponds to the WHO plausibility bounds for anthropometric data. This indicator serves as the primary dependent variable in subsequent regression analyses examining systematic patterns of anthropometric missingness.

For the continuous-outcome models, children with missing HAZ and those with biologically implausible HAZ ($|Z| > 6$) were treated as having missing HAZ. Multiple imputation was then used to address missingness in HAZ and selected covariates under a Missing at Random (MAR) assumption, conditional on the observed predictors included in the imputation model. Results from imputed models are interpreted as a robustness check for the stability of substantive inferences relative to complete-case analysis, rather than as definitive evidence about the true unobserved values.

3.4 Explanatory Variables

A comprehensive set of child, maternal, and household level covariates was included to examine potential predictors of height usability.

Child-level variables include age in months (continuous), sex (1 = female), recent illness (indicator equal to 1 if the child experienced diarrhea, fever, or symptoms of acute respiratory infection within the preceding two weeks), and the number of children under five residing in the household. Age was modeled continuously to capture fine grained variation in anthropometric measurement patterns.

Maternal characteristics include mother’s age (continuous), educational attainment (four categories: no education, primary, secondary, higher) and pregnancy status at the time of survey (1 = pregnant). Educational attainment is commonly used as a proxy for socioeconomic position and knowledge related to maternal health, both of which may influence the completeness of the anthropometric measurement.

Household characteristics include the wealth quintile (five categories: poorest, poorer, middle, richer, and richest), the sex of the head of the household (1 = female), and the place of residence (1 = rural). Three observations with ambiguous coding in the household head variable were excluded due to negligible frequency and lack of analytical interpretability.

To assess potential temporal variation in anthropometric data completeness, surveys were grouped into two periods: 2011–2016 and 2017–2022. A binary time indicator was generated to facilitate interaction analyses across survey waves. The temporal division reflects approximate mid-point separation of survey rounds to examine potential changes in measurement completeness across earlier and later DHS waves.

3.5 Analytical Sample

The pooled dataset included all children aged 0–59 months from eleven DHS survey rounds conducted between 2011 and 2022, yielding 566,663 eligible children in the unweighted data. Sample contributions varied across countries, with India contributing the majority (492,332), followed by Bangladesh (34,182), Pakistan (24,433), and Nepal (15,716). After applying DHS sampling weights, the pooled population-referenced total was 552,986 children. Because country sample sizes differ substantially, pooled descriptive estimates reflect the combined DHS sample rather than equal-country or regionally population-weighted totals. A detailed breakdown by country and survey round is provided in Appendix Table A.1.

Following survey declaration, the full estimation sample comprised $N = 564,428$, with minor differences from the raw unweighted count arising from survey-design adjustments, including the omission of strata containing no population members. All descriptive prevalence estimates of height usability were calculated using this survey-declared estimation sample.

For survey-weighted logistic regression models examining determinants of height usability, all children were retained regardless of anthropometric usability status, since missing and implausible height values constitute the outcome. Observations with missing values on explanatory covariates were excluded, resulting in a final analytic sample of 509,006 observations.

For linear regression models predicting HAZ, complete-case analyses were restricted to children with usable anthropometric measurements and complete covariate information. In multiple imputation analyses, unusable HAZ values and missing child age were imputed using Predictive Mean Matching (number of imputations, $m = 20$), and estimates were pooled using Rubin’s rules. Consequently, sample sizes vary across descriptive tables and regression models due to differences in survey declaration, outcome definition, covariate completeness, and imputation procedures; these differences are methodologically appropriate under complex survey design and do not reflect data inconsistencies.

Chapter 4

Results

All statistical analyses were conducted using Stata version 17.0 and R version 4.3.1. Stata was used for survey-weighted descriptive analysis, bivariate comparisons, logistic regression modeling, and selected graphical outputs. R was used for data cleaning, multiple imputation, and additional data visualization using the packages `mice`, `tidyverse`, `haven`, `scales`, `survey` and `lattice`.

4.1 Prevalence of Usable Height Measurements

Figure 4.1 summarizes the overall availability of usable height-for-age information in pooled DHS surveys conducted between 2011 and 2022. Across the pooled sample, approximately 82.8% of children had usable height measurements, while 17.2% were classified as unusable (missing or implausible). This overall prevalence provides the baseline context for assessing whether anthropometric missingness is randomly distributed or systematically patterned across observed characteristics.

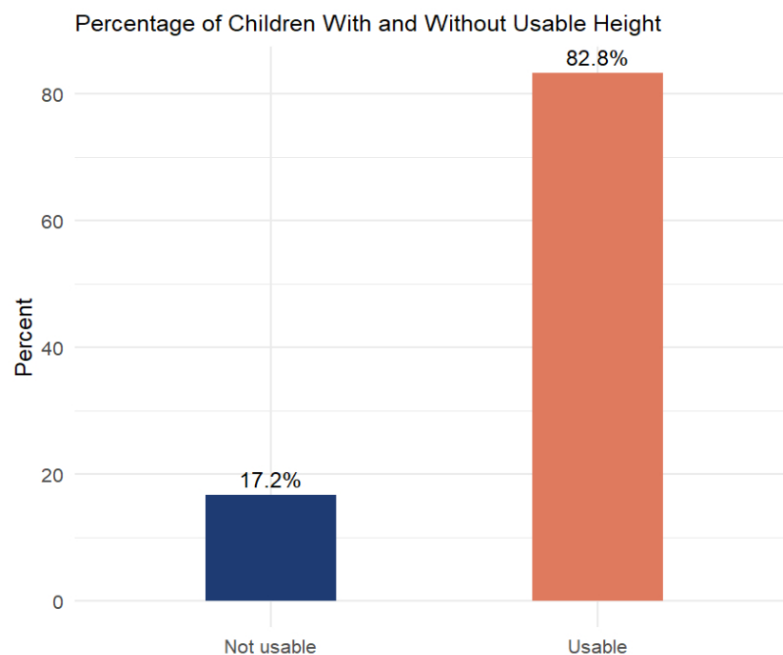


Figure 4.1: Percentage of children with and without usable height measurements in pooled DHS surveys (2011–2022).

Table 4.1 compares child, maternal, and household characteristics between children with usable and unusable height data. Systematic differences are evident across multiple domains. Children with usable measurements were, on average, older than those with unusable measurements, suggesting that measurement completeness varies by child age. Several socioeconomic indicators also differed between groups, including maternal education and household wealth. Rural residence was more common among children with usable height data than among those without usable measurements, indicating that usability patterns are not purely driven by urban–rural access constraints. The presence of statistically significant bivariate differences suggests that the assumption of MCAR is inconsistent in this pooled dataset, motivating the multivariable analyses presented in subsequent sections.

Table 4.1: Background characteristics of children by height usability status in pooled DHS surveys (2011–2022)

Characteristic	Usable height (n = 457,841)	Not usable height (n = 95,145)	p-value
Child characteristics			
Child age, months (mean $\pm$ SD)	29.90 $\pm$ 17.20	26.62 $\pm$ 17.55	<0.001
Female sex, n (%)	220,594 (48.18)	44,757 (47.04)	<0.001
Ill in past 2 weeks, n (%)	113,208 (24.73)	20,322 (21.36)	<0.001
Maternal characteristics			
Mother’s age, years (mean $\pm$ SD)	26.84 $\pm$ 5.01	27.16 $\pm$ 5.33	<0.001
No education, n (%)	115,865 (25.31)	30,303 (31.85)	
Primary education, n (%)	65,526 (14.31)	15,285 (16.07)	
Secondary education, n (%)	218,212 (47.66)	37,887 (39.82)	
Higher education, n (%)	58,238 (12.72)	11,669 (12.26)	<0.001
Currently pregnant, n (%)	33,504 (7.32)	8,714 (9.16)	<0.001
Household characteristics			
Poorest, n (%)	112,331 (24.54)	24,449 (25.60)	
Poorer, n (%)	99,976 (21.84)	20,021 (21.04)	
Middle, n (%)	90,734 (19.82)	18,407 (19.35)	
Richer, n (%)	84,922 (18.55)	16,790 (17.65)	
Richest, n (%)	69,879 (15.26)	15,478 (16.26)	<0.001
Female household head, n (%)	62,584 (13.67)	13,382 (14.06)	0.049
Children under five in household (mean $\pm$ SD)	1.70 $\pm$ 0.94	1.58 $\pm$ 1.11	<0.001
Rural residence, n (%)	333,248 (72.79)	65,851 (69.21)	<0.001

Notes: Values are survey-weighted means $\pm$ standard deviation (SD) for continuous variables and number (percentage) for categorical variables. P-values are derived from survey-weighted t-tests (continuous variables) and survey-weighted chi-square tests (categorical variables). Estimates account for DHS sampling weights, clustering at the PSU level, and stratification.

4.2 Age Pattern of Height Measurement Usability

Figure 4.2 displays the distribution of child age in months stratified by height usability status. A clear divergence in age patterns is observed between children with usable and unusable height measurements. The density of unusable observations is disproportionately concentrated among younger children, particularly those below 24 months of age.

In contrast, children with usable height data are more evenly distributed across the age range, with a noticeable shift toward older ages. This pattern suggests that anthropometric measurement completeness improves as children grow older. The higher prevalence of unusable measurements among younger children

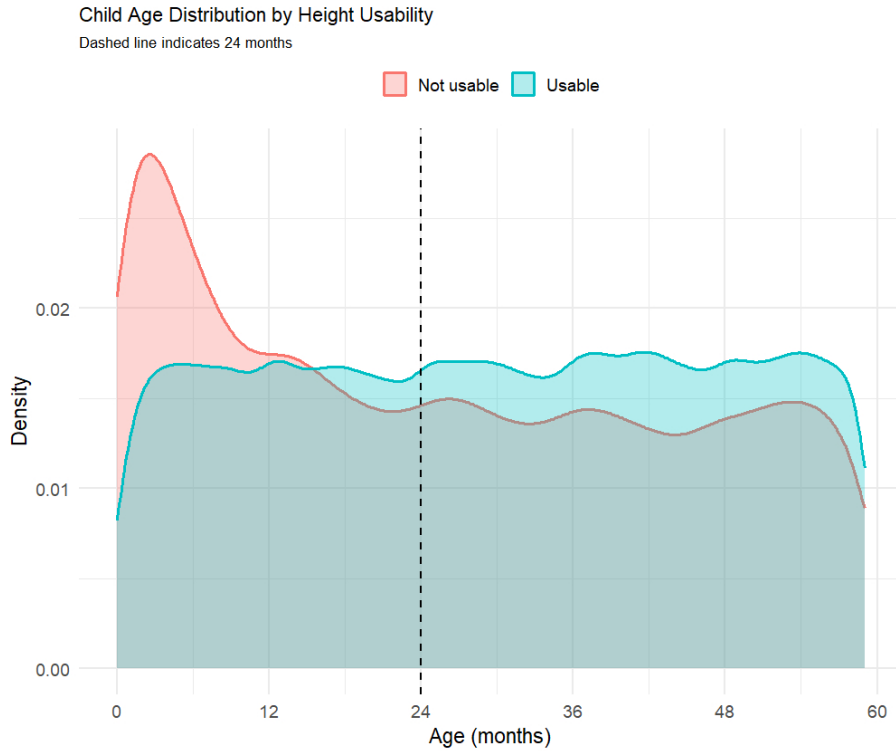


Figure 4.2: Distribution of child age (in months) by height usability status in pooled DHS surveys (2011–2022).

is consistent with documented operational challenges in obtaining accurate length measurements for infants.

These findings indicate that age is a critical determinant of height data usability and reinforce the descriptive results presented in Table 4.1, where children with usable measurements were, on average, older than those without usable measurements. The pronounced age gradient further supports the need to account for age in multivariable models assessing systematic patterns of anthropometric missingness. Additional descriptive statistics stratified by child age category (<24 vs ≥ 24 months) are presented in Appendix Table A.2, which further illustrates the concentration of unusable measurements among younger children.

4.3 Determinants of Height Measurement Usability

Table 4.2 presents results from the survey-weighted logistic regression model estimating adjusted odds ratios for height usability in pooled DHS surveys. The model includes child, maternal, and household level characteristics.

At the child level, age in months is positively associated with usable height measurements. Each additional month of age slightly increases the odds of having a usable HAZ value. Female children exhibit higher odds of usable height data compared to male children. Recent illness in the past two weeks is also associated with higher odds of usability, while a greater number of children under five in the household is associated with lower odds of obtaining usable measurements.

Among maternal characteristics, primary and secondary education are significantly associated with higher odds of height usability relative to no formal education. However, higher education does not

Table 4.2: Survey-weighted logistic regression predicting usable child height data (pooled DHS surveys, 2011–2022)

Variable	Level	Adj. OR	95% CI	p-value
Child characteristics				
Child age (months)	Continuous	1.012	(1.011, 1.013)	< 0.001
Sex of child	Male	1.00		
	Female	1.079	(1.046, 1.115)	< 0.001
Ill in past 2 weeks	No	1.00		
	Yes	1.340	(1.287, 1.395)	< 0.001
Number of children in household	Continuous	0.966	(0.948, 0.985)	0.001
Maternal characteristics				
Mother's age (years)	Continuous	0.999	(0.995, 1.003)	0.675
Mother's education	No education	1.00		
	Primary	1.162	(1.098, 1.229)	< 0.001
	Secondary	1.164	(1.111, 1.219)	< 0.001
	Higher	1.011	(0.944, 1.083)	0.759
Currently pregnant	No	1.00		
	Yes	1.169	(1.092, 1.250)	< 0.001
Household characteristics				
Wealth index	Poorest	1.00		
	Poorer	1.052	(1.003, 1.104)	0.038
	Middle	1.005	(0.952, 1.061)	0.858
	Richer	0.979	(0.915, 1.049)	0.547
	Richest	0.776	(0.726, 0.830)	< 0.001
Household head	Male	1.00		
	Female	1.039	(0.989, 1.092)	0.133
Residence	Urban	1.00		
	Rural	1.344	(1.274, 1.417)	< 0.001

Model information. Observations: 509,006. PSUs (clusters): 62,643. Outcome: usable height data (1 = usable; 0 = missing/improbable). Odds ratios are derived from survey-weighted logistic regression models accounting for DHS sampling weights, clustering at the PSU level, and stratification.

show a statistically significant association. Maternal age is not significantly related to usability after adjustment. Children of mothers who were pregnant at the time of survey have higher odds of usable height measurements.

At the household level, compared to the poorest households, children in the richest quintile have significantly lower odds of having usable height measurements ($OR < 1$), indicating that the association between wealth and usability is non-monotonic in the pooled model, with the richest group showing lower odds of usable measurement than the poorest, suggesting that the pooled association does not follow a simple linear gradient across wealth categories. Rural residence is strongly associated with higher odds of usable height data compared to urban residence. In contrast, the sex of the household head does not show a statistically significant association.

To assess whether the observed associations were driven by country-specific or survey-period effects, an extended pooled model including country and survey period indicators was estimated. Substantial cross-country variation in the prevalence of non-usable height measurements is illustrated in Appendix Figure A.1. However, when country and survey-period fixed effects are included (Appendix Table A.5), the magnitude and statistical significance of the primary covariates remain substantively unchanged. This indicates that the associations between height usability and demographic and socioeconomic characteristics are not merely artifacts of particular countries or survey waves.

4.4 Temporal Variation in Height Usability

Table 4.3: Adjusted odds ratios for availability of usable height data, stratified by survey period

Covariate	Level	2011–2016		2017–2022	
		Adj. OR	95% CI	Adj. OR	95% CI
Child characteristics					
Child age (months)	Continuous	1.01	(1.01, 1.02)	1.01	(1.01, 1.01)
Sex of child	Male	1.00		1.00	
	Female	1.12	(1.07, 1.17)	1.04	(0.99, 1.08)
Ill in past 2 weeks	No	1.00		1.00	
	Yes	1.34	(1.28, 1.42)	1.34	(1.26, 1.42)
Number of children in household	Continuous	0.97	(0.95, 0.99)	0.96	(0.93, 0.99)
Mother’s age (years)	Continuous	1.00	(0.99, 1.00)	1.00	(0.99, 1.00)
Mother’s education	No education	1.00		1.00	
	Primary	1.15	(1.06, 1.24)	1.16	(1.07, 1.27)
	Secondary	1.14	(1.07, 1.21)	1.15	(1.07, 1.23)
	Higher	0.99	(0.90, 1.10)	0.97	(0.89, 1.07)
Currently pregnant	No	1.00		1.00	
	Yes	1.23	(1.12, 1.35)	1.10	(1.00, 1.21)
Wealth index	Poorest	1.00		1.00	
	Poorer	1.04	(0.98, 1.11)	1.07	(1.00, 1.15)
	Middle	0.98	(0.91, 1.05)	1.06	(0.97, 1.15)
	Richer	0.95	(0.86, 1.05)	1.03	(0.94, 1.13)
	Richest	0.81	(0.74, 0.89)	0.76	(0.69, 0.84)
Household head	Male	1.00		1.00	
	Female	1.07	(1.00, 1.15)	1.00	(0.93, 1.07)
Residence	Urban	1.00		1.00	
	Rural	1.30	(1.21, 1.40)	1.38	(1.28, 1.49)
Model information					
Observations			268,954		240,052
PSUs (clusters)			30,658		31,985
Percentage without usable height			17.08%		16.32%

Notes: Odds ratios (OR) are derived from survey-weighted logistic regression models accounting for DHS sampling weights, clustering at the primary sampling unit (PSU) level, and stratification (linearized standard errors). The outcome is availability of usable height data (1 = usable; 0 = missing/implausible).

To examine whether the determinants of height measurement usability changed over time, survey-weighted logistic regression models were estimated separately for the periods 2011–2016 and 2017–2022. The stratified results, presented in Table 4.3, allow comparison of the magnitude and direction of associations across survey waves.

Across both time periods, child age remains positively associated with height usability, indicating that older children consistently exhibit higher odds of having usable measurements. The negative association between the number of children under five in the household and usability also persists across waves, suggesting that household-level resource constraints continue to influence measurement completeness over time. These stable patterns indicate that core demographic determinants of anthropometric usability remain largely unchanged.

Maternal education continues to display a positive relationship with usability in both periods, although

the strength of association varies modestly between waves. Primary and secondary education remain significantly associated with higher odds of usable height data relative to no education. In contrast, the association between household wealth and usability remains non-monotonic across survey periods. While children in the “poorer” category sometimes exhibit slightly higher odds relative to the poorest group, the richest quintile consistently demonstrates lower odds of usable height measurements. The overall structure of the wealth–usability relationship does not reverse over time, although the magnitude of differences varies modestly between survey periods.

Geographic differences also persist across waves. Rural residence remains positively associated with height usability in both periods, with a somewhat stronger association observed in the later surveys. Overall, the stratified analysis indicates that although the fundamental determinants of anthropometric usability remain stable, the strength of certain socioeconomic and geographic gradients evolves over time. These findings provide the basis for formally testing interaction effects between survey period and key socioeconomic variables.

4.5 Interaction Effects Between Survey Period and Socioeconomic Factors

To assess whether socioeconomic disparities in height usability changed over time, interaction terms between survey period and key household characteristics were incorporated into the survey-weighted logistic regression models. The interaction between wealth quintile and survey period was statistically significant ($p\text{-value} = 0.0487$), whereas the interaction between residence and survey period did not reach conventional levels of statistical significance ($p\text{-value} = 0.0844$). Predicted probabilities are estimated from the regression analytic sample $N = 509,006$, which excludes observations with missing child age, and therefore differs from the full pooled descriptive sample.

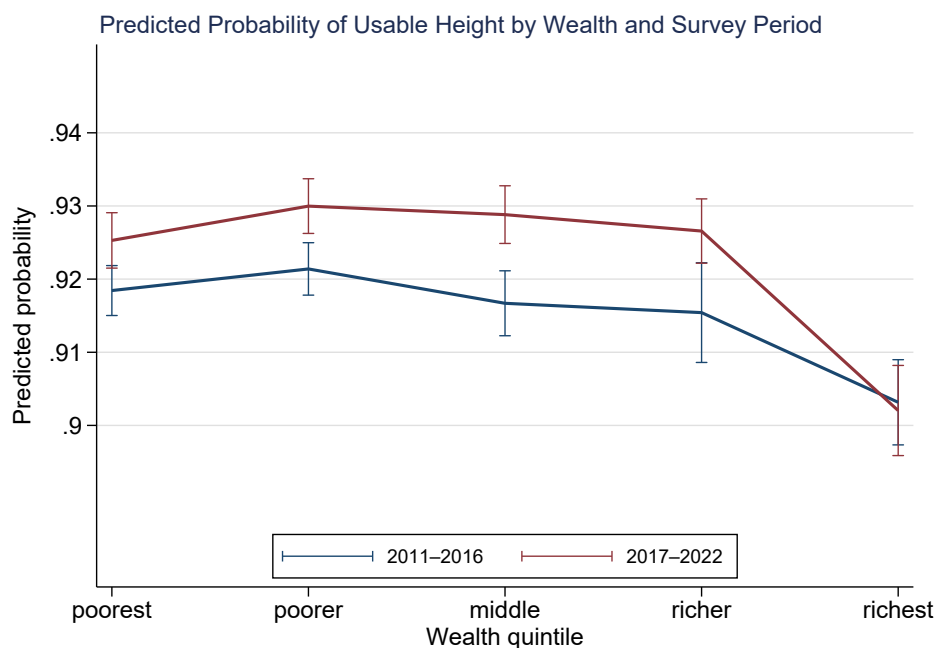


Figure 4.3: Predicted probability of usable height measurement by wealth quintile and survey period.

Figure 4.3 presents predicted probabilities of usable height measurements across wealth quintiles for the periods 2011–2016 and 2017–2022. Across all wealth categories, predicted usability is modestly higher in the later survey period, indicating an overall improvement in anthropometric measurement completeness over time. However, the pattern across wealth quintiles remains distinctly non-monotonic in both periods. Predicted probabilities increase slightly from the poorest to the poorer and middle quintiles, decline among the richer group, and drop more sharply for the richest quintile.

Importantly, children in the richest households consistently exhibit the lowest predicted probability of usable height measurement in both survey periods. Although the wealth and period interaction is statistically significant, the figure indicates that temporal change primarily reflects a general upward shift in usability rather than a fundamental alteration of the wealth–usability structure. The non-linear pattern particularly the persistent reduction in usability at the highest wealth level remains stable across survey waves. The corresponding predicted probabilities for wealth quantiles are presented in Table A.3.

Predicted probabilities for residence are presented in Table 4.4. Rural children demonstrate higher predicted usability than urban children in both periods. While overall usability increases slightly over time, the rural–urban gap remains relatively constant. The absence of a statistically significant interaction effect suggests that geographic disparities in anthropometric usability persist but do not meaningfully expand or contract across survey waves. The corresponding interaction plot is shown in Appendix Figure A.2.

Table 4.4: Predicted probability of usable height data by residence and survey period

Survey period	Urban	Rural	Rural–Urban difference
2011–2016	0.901 (0.896, 0.907)	0.922 (0.920, 0.924)	0.021
2017–2022	0.905 (0.900, 0.910)	0.930 (0.928, 0.932)	0.025

Notes: Predicted probabilities are marginal estimates from a survey-weighted logistic regression including a residence $\times$ survey period interaction. Values are predicted probabilities with 95% confidence intervals. Joint Wald test for interaction: $p = 0.0844$.

Taken together, the interaction analyses indicate that while overall height measurement usability improved modestly in later survey waves, temporal changes were not uniform across socioeconomic groups. Wealth-related disparities exhibit statistically detectable shifts over time, whereas rural–urban differences remain comparatively stable.

4.6 Determinants of Height-for-Age Z-score: Complete-Case and Multiple Imputation Analysis

Before estimating the HAZ regression models, Appendix Table A.4 summarizes survey-weighted mean HAZ ($\pm$ SD) by key child, maternal, and household characteristics. Mean HAZ varies systematically across socioeconomic strata, with higher values observed among children from wealthier households and those with more educated mothers. These descriptive patterns provide context for the multivariable results reported below.

Table 4.5 presents survey-weighted linear regression estimates for height-for-age Z-scores (HAZ) using both complete-case analysis and multiple imputation with predictive mean matching. The comparison allows assessment of whether systematic patterns of missing anthropometric data meaningfully alter substantive conclusions.

Table 4.5: Determinants of Height-for-Age Z-score (HAZ): Complete-case versus Multiple Imputation

Variables	Complete case			Multiple Imputation (PMM)		
	Adj. β	95% CI	p-value	Adj. β	95% CI	p-value
Child age (months)	-0.016	(-0.016, -0.016)	<0.001	-0.016	(-0.017, -0.016)	<0.001
Female child	0.080	(0.066, 0.094)	<0.001	0.080	(0.067, 0.093)	<0.001
Child illness (yes)	-0.079	(-0.095, -0.063)	<0.001	-0.075	(-0.090, -0.060)	<0.001
Children under five in HH	-0.086	(-0.094, -0.078)	<0.001	-0.085	(-0.092, -0.078)	<0.001
Mother's age (years)	0.008	(0.006, 0.009)	<0.001	0.009	(0.008, 0.010)	<0.001
Mother's education (ref: no education)						
Primary	0.122	(0.099, 0.145)	<0.001	0.115	(0.095, 0.135)	<0.001
Secondary	0.303	(0.284, 0.323)	<0.001	0.303	(0.285, 0.320)	<0.001
Higher	0.533	(0.503, 0.563)	<0.001	0.532	(0.502, 0.561)	<0.001
Currently pregnant (yes)	-0.235	(-0.260, -0.210)	<0.001	-0.217	(-0.242, -0.191)	<0.001
Wealth index (ref: poorest)						
Poorer	0.171	(0.151, 0.191)	<0.001	0.170	(0.152, 0.188)	<0.001
Middle	0.312	(0.290, 0.335)	<0.001	0.317	(0.297, 0.338)	<0.001
Richer	0.483	(0.457, 0.509)	<0.001	0.481	(0.457, 0.506)	<0.001
Richest	0.682	(0.650, 0.713)	<0.001	0.675	(0.647, 0.703)	<0.001
Rural residence	0.016	(-0.007, 0.039)	0.178	0.014	(-0.008, 0.036)	0.222
Female household head	-0.009	(-0.030, 0.012)	0.402	-0.016	(-0.035, 0.002)	0.087

Notes: Survey-weighted linear regression; 95% CIs use linearized variance estimation. MI uses PMM ($m = 20$) with Rubin's rules; not usable HAZ scores and missing child age (months) were imputed. Reference categories: male child; no child illness; no maternal education; not pregnant; poorest; urban; male household head.

Across both approaches, the direction and magnitude of associations are remarkably consistent. Child age is negatively associated with HAZ, indicating that height-for-age declines modestly with increasing age in months. Female children exhibit slightly higher HAZ scores compared to males, while recent child illness is associated with lower HAZ values. The number of children under five in the household remains negatively associated with child height outcomes in both specifications.

Maternal education demonstrates a strong and graded association with HAZ in both models. Relative to no education, primary, secondary, and higher education are associated with progressively higher Z-scores. Children of mothers who were pregnant at the time of survey exhibit lower HAZ scores in both specifications. Wealth differences shows a strong positive gradient, with progressively higher HAZ scores observed across increasing wealth quintiles. Children in the richest households exhibit substantially higher HAZ values relative to the poorest group.

Importantly, the statistical significance and substantive magnitude of coefficients remain largely unchanged between complete-case and multiply imputed analyses. Variables that are non-significant in the complete-case model remain non-significant under imputation, and significant predictors retain similar effect sizes. This stability suggests that although anthropometric usability is systematically patterned, the exclusion of missing cases does not appear to materially alter estimated relationships between socioeconomic characteristics and child nutritional status. Thus, the close alignment between complete-case and multiply imputed results provides reassurance regarding the robustness of substantive inferences drawn from the HAZ regression models.

Supplementary diagnostic plots assessing the distribution of observed and imputed values, as well as convergence of the imputation algorithm, are presented in Appendix Figures A.3 – A.5.

4.7 Sensitivity Analysis: Excluding Child Age

As a robustness check, a survey-weighted logistic regression model was estimated excluding child age as a covariate. This specification assesses whether the observed associations between socioeconomic characteristics and height usability are driven primarily by age-related measurement patterns. Results are presented in Table 4.6.

Table 4.6: Sensitivity analysis: Survey-weighted logistic regression predicting usable child height excluding child age (full sample)

Covariate	Level	Adjusted OR	95% CI	p-value
Sex of child (ref: male)				
	Female	1.047	(1.026, 1.068)	<0.001
Ill in past 2 weeks (ref: no/unsure)				
	Yes	1.207	(1.174, 1.240)	<0.001
Number of children in household	Continuous	1.175	(1.152, 1.199)	<0.001
Mother's age (years)	Continuous	0.998	(0.996, 1.000)	0.091
Mother's education (ref: no education)				
	Primary	1.171	(1.129, 1.215)	<0.001
	Secondary	1.677	(1.621, 1.734)	<0.001
	Higher	1.653	(1.572, 1.739)	<0.001
Currently pregnant (ref: no/unsure)				
	Yes	0.806	(0.775, 0.839)	<0.001
Wealth index (ref: poorest)				
	Poorer	0.984	(0.951, 1.020)	0.391
	Middle	0.923	(0.884, 0.963)	<0.001
	Richer	0.933	(0.888, 0.981)	0.007
	Richest	0.833	(0.788, 0.880)	<0.001
Sex of household head (ref: male)				
	Female	0.981	(0.949, 1.014)	0.252
Residence (ref: urban)				
	Rural	1.215	(1.171, 1.262)	<0.001
Model information				
Observations				564,428
PSUs (clusters)				62,720

Notes: Odds ratios are derived from survey-weighted logistic regression models with linearized standard errors, accounting for DHS sampling weights, PSU clustering, and stratification.

The overall pattern of associations remains substantively consistent with the baseline model. Female children continue to exhibit higher odds of usable height measurements relative to males, and recent child illness remains positively associated with usability. In particular, the estimated association for household composition changes when child age is excluded. The odds ratio for the number of children under five shifts from below 1 in the baseline model Table 4.2 to above 1 in the no-age specification Table 4.6, indicating sensitivity of this covariate to model specification. This sign reversal suggests that child age acts as a confounder in the association between household composition and measurement usability. Because younger children are both more likely to have unusable measurements and more likely to reside in households with multiple young children, omitting age induces bias in the estimated household effect. Maternal current pregnancy status also shows sensitivity to the exclusion of child age. In the baseline model adjusting

for age Table 4.2, current pregnancy is positively associated with height usability, whereas in the no age specification Table 4.6 the association becomes negative. This sign reversal is consistent with confounding by child age and related household composition dynamics. Therefore, the pregnancy coefficient should be interpreted cautiously and primarily in the context of the fully adjusted specification.

Maternal education continues to display a strong and graded relationship with height usability. Relative to mothers with no education, primary, secondary, and higher education remain associated with higher odds of usable measurements. Surprisingly, wealth gradients remain pronounced, with children from richer households exhibiting lower usability compared to the poorest quintile. Rural residence continues to demonstrate a positive association with usability, consistent with earlier findings.

Although the exclusion of child age slightly alters coefficient magnitudes, the direction and statistical significance of key predictors remain unchanged. This stability indicates that the systematic patterns identified in the main model are not solely attributable to age-related measurement dynamics. The sensitivity analysis therefore reinforces the robustness of the baseline findings regarding socioeconomic and geographic disparities in anthropometric data usability.

Chapter 5

Discussion and Conclusion

5.1 Principal Findings and Discussion

This study provides a systematic evaluation of anthropometric measurement usability in pooled South Asian DHS surveys and demonstrates that missing or biologically implausible height-for-age observations are structured rather than randomly distributed. Across eleven survey rounds and four countries, unusable measurements were consistently associated with observable demographic and socioeconomic characteristics. These findings reinforce the importance of formally assessing missing data mechanisms before relying on complete-case analysis in applied nutritional research.

From the perspective of Rubin’s missing data framework, the empirical patterns observed here are inconsistent with the Missing Completely at Random (MCAR) assumption. The most robust and persistent determinant of usability is child age. Younger children were substantially more likely to have unusable height measurements, a pattern that remained stable across countries and survey periods. This age gradient aligns with established operational challenges in measuring recumbent length among infants and very young children ([Uliaszek and Kerr, 1999](#); [Pullum, 2013](#)). The persistence of this association across contexts suggests that age-related measurement difficulty represents a structural feature of anthropometric data collection rather than an incidental survey irregularity.

Socioeconomic differentials further illustrate the structured nature of anthropometric usability. Maternal education displayed a consistent positive association with measurement completeness, indicating that anthropometric data quality may be socially stratified. In contrast, the association between household wealth and usability was distinctly non-monotonic. Rather than exhibiting a linear gradient, predicted usability increased modestly from the poorest to middle quintiles but declined among the richest households, which consistently demonstrated the lowest predicted probability of usable measurement. This pattern challenges the assumption that greater socioeconomic advantage uniformly translates into improved survey participation or measurement completeness. Instead, it suggests that operational dynamics, availability constraints, or contextual factors in more affluent settings may shape anthropometric data collection in complex ways.

Temporal analyses reveal modest improvements in overall usability across survey waves. However, these improvements primarily reflect an upward shift in predicted usability rather than a structural transformation of underlying disparities. The non-linear wealth pattern, particularly the reduced usability among the richest quintile remained stable over time. Although the wealth and period interaction reached

statistical significance, visual and model-based evidence indicate that temporal change did not reverse or eliminate existing structural differences. Rural–urban disparities likewise persisted with minimal temporal expansion or contraction. These findings suggest that incremental improvements in survey implementation do not automatically resolve patterned data incompleteness.

Importantly, the presence of structured missingness did not translate into substantial distortion of substantive nutritional inference. Comparisons between complete-case and multiply imputed regression models for height-for-age Z-scores revealed remarkable stability in coefficient magnitudes and statistical significance. Conditional on observed covariates, exclusion of unusable cases did not materially alter estimated socioeconomic gradients in child growth. This distinction is methodologically important: missingness can be socially patterned without necessarily inducing large inferential bias when key predictors are appropriately specified.

Taken together, the findings conceptualize anthropometric missingness not as random noise, but as an embedded component of the data-generating process in complex household surveys. The contribution of this study lies not in questioning the reliability of DHS anthropometric data, but in demonstrating that systematic evaluation of missingness mechanisms is a prerequisite for rigorous secondary analysis. Explicit diagnostic assessment enhances methodological transparency and strengthens the credibility of population-level nutritional inference.

These findings also motivate several practical recommendations for secondary analyses using DHS anthropometry. Researchers should routinely report usability rates overall and by key subgroups, particularly child age and socioeconomic position, rather than treating missingness as a routine data processing decision. In outcome models of child growth, analysts should document the analytic sample restriction induced by unusable measurements and, where feasible, assess sensitivity to principled missing data approaches such as multiple imputation under plausible MAR assumptions. Finally, because usability patterns may vary across national implementations and survey rounds, presenting country or round-specific diagnostics alongside pooled results can help distinguish general structural patterns from context-specific fieldwork dynamics, thereby increasing analytic transparency and reducing ambiguity in interpretation.

5.2 Limitations

Several limitations should be considered when interpreting the findings of this study. First, the analysis relies on repeated cross-sectional survey data, which precludes causal inference regarding the mechanisms underlying anthropometric missingness. The associations identified reflect systematic patterns but do not establish the behavioral, logistical, or operational processes generating unusable measurements.

Second, although a broad set of child, maternal, and household level covariates was incorporated, unobserved factors, such as enumerator characteristics, supervision quality, fieldwork conditions, or community level implementation differences can also influence anthropometric usability. These factors are not directly captured in standard DHS recode files and therefore could not be explicitly modeled.

Third, the pooled dataset combines eleven DHS survey rounds conducted across different countries and time periods. Although harmonization procedures were applied and survey-design features were reconstructed to preserve design-based inference, differences in national implementation practices, training protocols, field logistics, and contextual conditions may contribute to cross-country heterogeneity in data quality. The prevalence of missingness and the completeness of the measurement are not uniform between

countries or survey waves, and such heterogeneity may introduce residual variability that cannot be fully accounted for in pooled models.

Fourth, missingness in child age required sample restrictions in regression analyses. In this study, child age in months was taken from the DHS anthropometry module variable `hw1`, which may be unrecorded when anthropometric measurement is not completed. Consequently, models that adjust for child age are estimated on a covariate complete subset, and analytic sample sizes differ across descriptive summaries, usability (logistic) models, and HAZ regressions. In addition, adjustments for single-unit strata in some survey rounds produced minor differences in effective sample size across specifications. Although these procedures are methodologically appropriate, they may limit generalizability of regression based results to the full pooled sample and can complicate direct comparison of sample counts across tables.

Fifth, the classification of biologically implausible values follows the World Health Organization recommended plausibility range for height-for-age Z-scores. Although widely accepted, alternative data-cleaning criteria could produce modest differences in estimates of usability prevalence.

Finally, multiple imputation was implemented under a Missing at Random (MAR) framework. Although the similarity between complete-case and imputed estimates provides reassurance, the true missing data mechanism cannot be definitively verified. If missingness depends on unobserved characteristics beyond those included in the models, residual bias may remain.

5.3 Overall Conclusions

This report evaluated whether missing and biologically implausible height-for-age measurements in pooled South Asian DHS data occur randomly or follow systematic patterns across demographic and socioeconomic groups. Using survey-weighted regression models, interaction analyses, and multiple imputation techniques, the study demonstrates that anthropometric usability is structured rather than random, yet its impact on multivariable nutritional inference appears limited under the modeling strategies examined.

The findings highlight three central conclusions. First, measurement usability exhibits a strong and persistent age gradient, with younger children disproportionately represented among unusable observations. Second, socioeconomic differentials, particularly maternal education and household wealth are associated with structured variation in anthropometric completeness. Third, although usability improved modestly over time, temporal changes were uneven across socioeconomic strata. Importantly, despite the presence of systematic missingness, substantive inferences regarding determinants of child height-for-age remained largely stable when comparing complete-case and multiple imputation approaches.

These results carry methodological implications for researchers working with large-scale survey data. Anthropometric datasets are often treated as analytically complete after routine cleaning and plausibility checks. However, the present study demonstrates that missingness and implausible values may reflect underlying demographic and socioeconomic structures. Automatically excluding incomplete observations without evaluating the mechanism of missingness may therefore obscure patterned data processes. Careful diagnostic assessment of missing data mechanisms should be considered a standard component of secondary data analysis.

5.4 Future Research Directions

Future research may extend this work in several directions. First, incorporating variables at the enumerator or fieldwork-level could help clarify operational factors that contribute to the usability of the anthropometric data. Second, comparative analyses across additional regions would allow evaluation of whether similar missingness structures emerge outside South Asia. Third, methodological extensions, such as selection models or pattern-mixture models could further explore the robustness of nutritional inferences under alternative assumptions about missing data mechanisms.

More broadly, this study underscores the importance of transparency in the treatment of incomplete anthropometric observations. DHS surveys remain among the most rigorous and policy-relevant health data systems globally. By systematically evaluating the structure and implications of anthropometric usability, this thesis contributes to strengthening the methodological foundations of nutritional research using complex survey data. Ensuring that missingness is explicitly examined rather than implicitly ignored enhances both analytic rigor and the credibility of population health inference.

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Appendix A

Supplementary Materials

Table A.1: Pooled Sample Composition by Country and Survey Round

Country	DHS Rounds Included	Unweighted n	Weighted Total	Weighted (%)
Bangladesh	2011, 2014, 2017–18, 2022	34,182	34,492	6.24
India	2015–16, 2019–21	492,332	480,403	86.87
Nepal	2011, 2016, 2022	15,716	15,644	2.83
Pakistan	2012–13, 2017–18	24,433	22,446	4.06
Total		566,663	552,986	100.00

Notes: Unweighted totals reflect raw pooled counts of children aged 0–59 months. Weighted totals are based on survey-adjusted tabulations using DHS sampling weights (v005) rescaled by 1,000,000. Percentages are calculated using weighted totals.

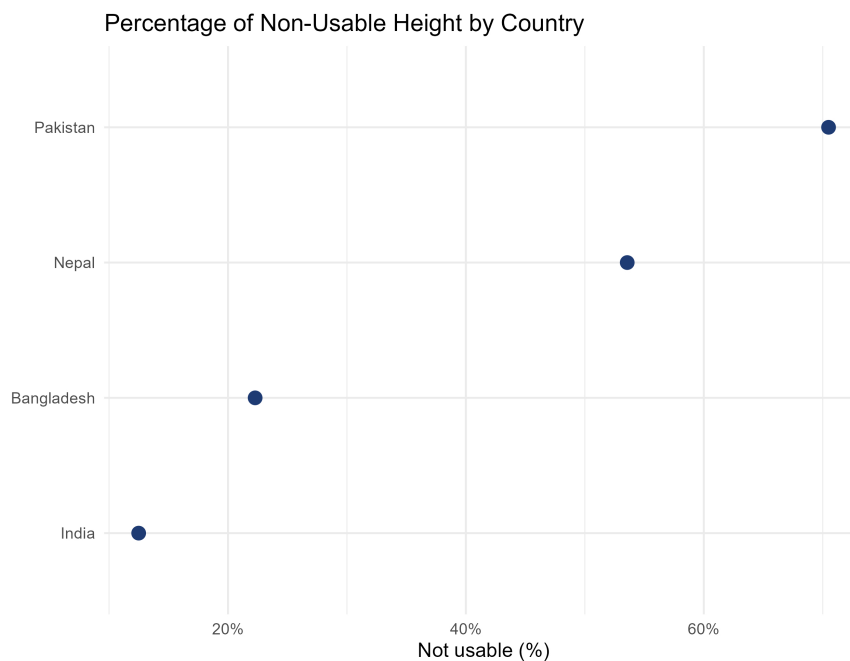


Figure A.1: Percentage of non-usable height measurements by country.

Table A.2: Descriptive characteristics stratified by child age (<24 vs ≥24 months) among children under five in pooled DHS surveys

Characteristic	All children (n = 552,986)	<24 months (n = 197,380)	≥24 months (n = 300,795)	p-value
Child characteristics				
Child age, months (mean ± SD)	29.64 ± 17.26	11.55 ± 6.78	41.50 ± 10.39	<0.001
Female sex, n (%)	265,351 (47.98)	94,961 (48.00)	144,317 (48.00)	0.519
Ill in past 2 weeks, n (%)	133,530 (24.15)	57,970 (29.40)	63,345 (21.10)	<0.001
Maternal characteristics				
Mother's age, years (mean ± SD)	26.89 ± 5.07	25.50 ± 4.77	27.72 ± 4.94	<0.001
No education, n (%)	146,168 (26.43)	44,954 (22.80)	81,087 (27.00)	
Primary education, n (%)	80,811 (14.62)	26,554 (13.50)	44,037 (14.60)	
Secondary education, n (%)	256,099 (46.31)	97,111 (49.20)	139,400 (46.30)	
Higher education, n (%)	69,907 (12.64)	28,762 (14.60)	36,271 (12.10)	<0.001
Currently pregnant, n (%)	42,218 (7.63)	11,041 (5.60)	24,791 (8.20)	<0.001
Household characteristics				
Poorest, n (%)	136,780 (24.73)	47,180 (23.90)	74,168 (24.70)	
Poorer, n (%)	119,997 (21.70)	42,697 (21.60)	64,857 (21.60)	
Middle, n (%)	109,141 (19.74)	39,701 (20.10)	58,450 (19.40)	
Richer, n (%)	101,711 (18.39)	36,790 (18.60)	55,733 (18.50)	
Richest, n (%)	85,357 (15.44)	31,013 (15.70)	47,586 (15.80)	<0.001
Female household head, n (%)	75,966 (13.74)	27,217 (13.80)	40,658 (13.60)	0.024
Number of children in household (mean ± SD)	1.68 ± 0.97	1.69 ± 0.97	1.71 ± 0.91	<0.001
Rural residence, n (%)	399,098 (72.17)	143,877 (72.90)	215,392 (71.60)	<0.001

Notes: Values are presented as mean ± standard deviation (SD) for continuous variables and number (percentage) for categorical variables. P-values are derived from survey-weighted t-tests for continuous variables and survey-weighted chi-square tests for categorical variables. Age category is missing for a subset of observations; <24 and ≥ 24 months are calculated among children with non-missing age.

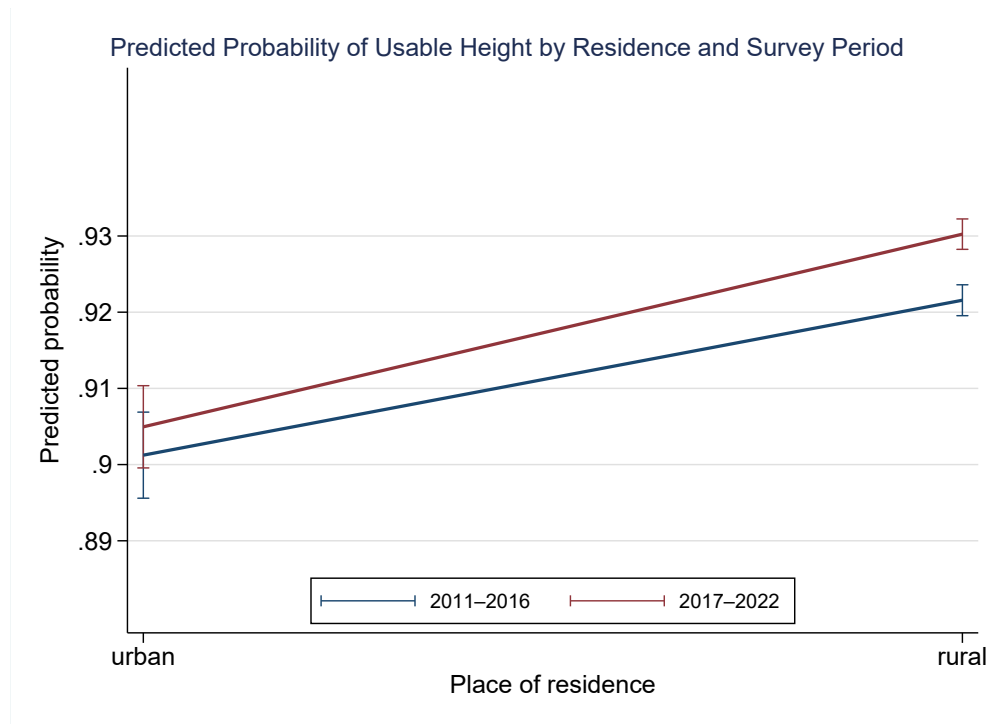


Figure A.2: Predicted probability of usable height by residence and survey period.

Table A.3: Predicted probability of usable height data by wealth quintile and survey period

Wealth quintile	2011–2016	2017–2022
Poorest	0.918 (0.915, 0.922)	0.925 (0.922, 0.929)
Poorer	0.921 (0.918, 0.925)	0.930 (0.926, 0.934)
Middle	0.917 (0.912, 0.921)	0.929 (0.925, 0.933)
Richer	0.915 (0.909, 0.922)	0.927 (0.922, 0.931)
Richest	0.903 (0.897, 0.909)	0.902 (0.896, 0.908)

Notes: Predicted probabilities are marginal estimates from a survey-weighted logistic regression including a wealth quintile $\times$ survey period interaction. Values are predicted probabilities with 95% confidence intervals. Joint Wald test for interaction: $p = 0.0487$.

Table A.4: Survey-weighted mean Height-for-Age Z-score (HAZ) by child, maternal, and household characteristics

Characteristic	Mean HAZ ($\pm$ SD)	p-value
Sex of child		
Male	-1.448 $\pm$ 1.705	
Female	-1.383 $\pm$ 1.716	<0.001
Ill in past 2 weeks		
No	-1.412 $\pm$ 1.731	
Yes	-1.431 $\pm$ 1.649	0.027
Mother's education		
No education	-1.836 $\pm$ 1.758	
Primary	-1.630 $\pm$ 1.650	
Secondary	-1.292 $\pm$ 1.661	
Higher	-0.812 $\pm$ 1.618	<0.001
Wealth index		
Poorest	-1.839 $\pm$ 1.784	
Poorer	-1.588 $\pm$ 1.728	
Middle	-1.384 $\pm$ 1.644	
Richer	-1.154 $\pm$ 1.578	
Richest	-0.857 $\pm$ 1.597	<0.001
Residence		
Urban	-1.173 $\pm$ 1.589	
Rural	-1.508 $\pm$ 1.744	<0.001

Notes: Values represent survey-weighted mean height-for-age z-scores (HAZ) with standard deviations. P-values are derived from survey-weighted linear regression models (Wald tests for categorical variables). Estimates account for sampling weights, clustering, and stratification.

Table A.5: Survey-weighted logistic regression predicting usable child height, adjusted for child, maternal, and household characteristics, country, and survey period

Variable	Adjusted OR	95% CI	p-value
Child age (months)	1.012	(1.011, 1.013)	<0.001
Child gender			
Female	1.078	(1.044, 1.113)	<0.001
Child illness, past two weeks			
Yes	1.326	(1.273, 1.381)	<0.001
Children under 5 in household	0.977	(0.958, 0.997)	0.024
Mother's age (years)	1.000	(0.996, 1.004)	0.997
Mother's education			
Primary	1.127	(1.065, 1.193)	<0.001
Secondary	1.147	(1.095, 1.202)	<0.001
Higher	0.993	(0.927, 1.064)	0.851
Currently pregnant			
Yes	1.191	(1.113, 1.274)	<0.001
Wealth index			
Poorer	1.056	(1.006, 1.108)	0.026
Middle	1.009	(0.956, 1.066)	0.738
Richer	0.986	(0.921, 1.056)	0.696
Richest	0.782	(0.730, 0.837)	<0.001
Sex of household head			
Female	1.014	(0.965, 1.066)	0.581
Residence type			
Rural	1.349	(1.279, 1.423)	<0.001
Country (ref: Bangladesh)			
India	0.797	(0.742, 0.855)	<0.001
Nepal	3.592	(2.832, 4.556)	<0.001
Pakistan	0.572	(0.503, 0.651)	<0.001
Survey period (ref: 2011–2016)			
2017–2022	1.114	(1.070, 1.160)	<0.001

Notes: Odds ratios are from a survey-weighted logistic regression with linearized standard errors, accounting for DHS sampling weights, PSU clustering, and stratification. The outcome is availability of usable child height data (1 = usable; 0 = missing/implausible).

Key R Code for Multiple Imputation

```
## Used packages
library(mice)
library(dplyr)
library(haven)
library(lattice)

## Data import from stata
DHS_pooled <- read_dta("DHS_pooled.dta")

## Select variables for imputation
imp_vars <- c(
  "haz",
  "child_age_months",
  "orig_mother_edu",
  "child_gender",
  "orig_wealth_index",
  "currently_pregnant",
  "child_illness",
  "residence_type",
  "children_under_5_in_hh",
  "mother_age",
  "sex_of_hh_head",
  "stratum_id",
  "psu_id",
  "sw"
)

imp_data <- DHS_pooled %>%
  select(all_of(imp_vars))

## Impute only continuous variables using PMM
meth <- make.method(imp_data)
meth[] <- "" # no imputation
meth["haz"] <- "pmm"
meth["child_age_months"] <- "pmm"

pred <- make.predictorMatrix(imp_data)

## Not imputing the design variables
design_vars <- c("stratum_id", "psu_id", "sw")
pred[design_vars, ] <- 0 # do not impute design variables.
pred[, design_vars] <- 0
pred["haz", "child_age_months"] <- 1
pred["child_age_months", "haz"] <- 1
pred

## Imputation vs observed plots
```

```

set.seed(123)

imp_pmm <- mice(
  imp_data,
  m = 20,
  method = meth,
  predictorMatrix = pred,
  maxit = 20,
  pmm.k = 5,
  seed = 123
)
plot(imp_pmm)

p1 <- densityplot(
  imp_pmm,
  ~ haz,
  main = list(
    "Observed vs Imputed HAZ Distribution\n(Predictive Mean Matching, m =
      20)",
    cex = 0.9
  ),
  xlab = "Height-for-Age Z-score (HAZ)",
  ylab = "Density",
  auto.key = list(space = "bottom"),
  par.settings = list(
    layout.widths = list(left.padding = 2, right.padding = 2)
  )
)
p1

p2 <- densityplot(
  imp_pmm,
  ~ child_age_months,
  main = list(
    "Observed vs Imputed Child Age (months) Distribution\n(Predictive Mean
      Matching, m = 20)",
    cex = 0.9
  ),
  xlab = "Child Age (months)",
  ylab = "Density")
p2

```

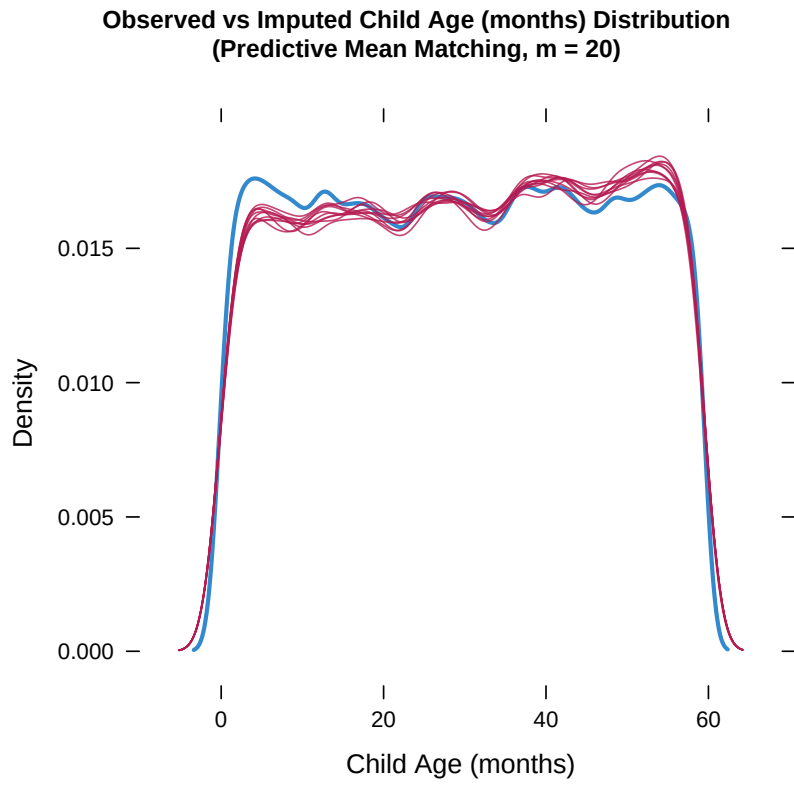


Figure A.3: Observed vs imputed distribution of child age.

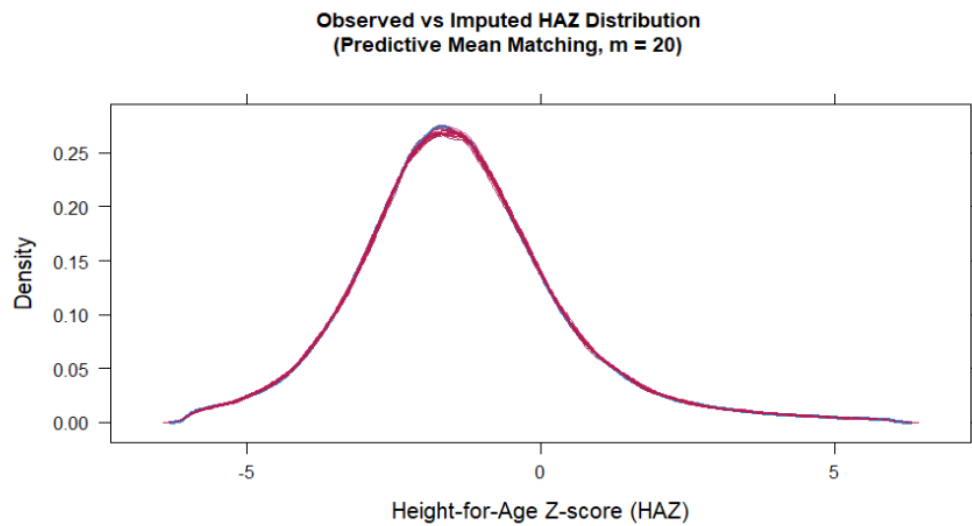


Figure A.4: Observed vs imputed distribution of HAZ score.

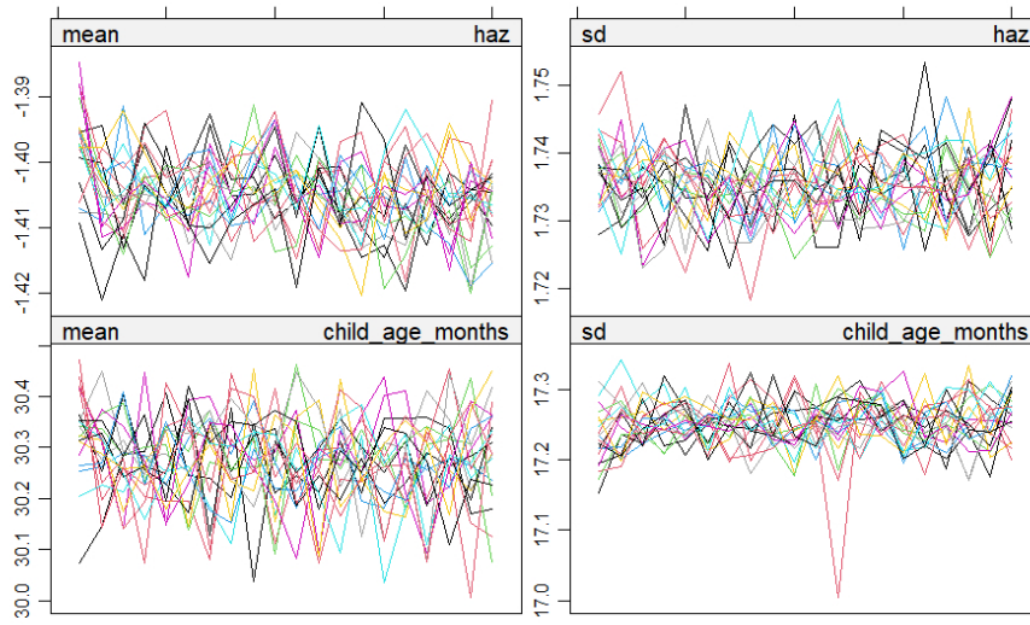


Figure A.5: Trace plot of multiple imputation iterations.

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